Week 2: Logistic Functions and Bounded Growth

Theme: "When Growth Has Limits"

Science Lesson: Fish populations, carrying capacity, sustainable harvesting, the Schaefer model

Mathematics Lesson: By the end of the week you should be able to:

1. Understand why exponential growth cannot continue indefinitely in real systems 2. Recognize and work with the logistic function as a model of bounded growth 3. Understand the Schaefer model for fish populations Science 4. Calculate Maximum Sustainable Yield (MSY) for renewable resources 5. Understand arithmetic sequences and series Mathematics 6. Understand Series Science 7. Compute Sums and Series Mathematics

Exam Alignment: QA, QUA, QU

2.1 Introduction: Why Growth Has Limits

The Problem with Unbounded Growth

In Week 1 we modeled exponential growth: $P(t) = P_0 e^{rt}$. This model captures the early stage of population growth beautifully. But nothing in a petri dish, outside an aquarium, or planet population is as sustainable as this curve suggests.

But here's the critical question: **Can our population grow sustainably forever?**

The answer is clearly: **no**. Consider:

Systems	Limiting Factor
Bacteria in petri dish	Nutrients exhausted, toxic accumulation
Fish in lake	Food supply, habitat, predators, disease
Human population	Resources, space, carrying capacity of Earth
Market for a product	Market saturation, competing products

Key Insight: All real growth systems eventually encounter **compounding limitations** on their sustainable growth. The exponential model is only valid when resources are abundant and constraints are far away.

From Week 1: Fish as Renewable Resource Management

When analyzing sustainable harvesting in Week 1, we assumed a fish stock is managed so:

- **Growth** (P_0): The total biomass at a point in time (e.g., 2 million kg)
- **Flow or Growth** (r): The change in stock over time (e.g., 400,000 tonnes of additional fish per year)

The fundamental question becomes: **How does the flow (growth) depend on the level of the stock?**

2.2 The Logistic Function

Building Intuition

A fish pond or a resource can't grow regardless of population size:

$$G(0) = 0$$

Then the stock could grow exponentially: $G(1) = 0.2\text{P}$

But as the stock grows, **competition for resources increases**. The growth rate slows down. This is the logistic function, characterized by a parameter called **carrying capacity** K.

The Logistic Growth Model

The logistic function incorporates this constraint elegantly:

$$P(t) = \frac{K}{1 + \frac{K - P_0}{P_0} e^{-rt}}$$

where: K = **carrying capacity** (maximum sustainable population) P_0 = initial population r = growth rate parameter t = time e = exponential constant (Euler's number, $e \approx 2.718$)

Alternative Form

The logistic function is sometimes written with an inflection point parameter t_0:

$$P(t) = \frac{K}{1 + e^{-r(t - t_0)}}$$

where: t_0 = carrying capacity, K = growth rate parameter t_0 = time when population reaches half of carrying capacity

Properties of the Logistic Function

Property	Value
Initial value	$P(0) = P_0$
Upper bound	$\lim_{t \to \infty} P(t) = K$
Midpoint	$P(t_0) = \frac{K}{2}$
Growth rate	$P'(t) = rP\left(1 - \frac{P}{K}\right)$
Shape	Symmetric about t_0
Behavior	Slow at first, accelerates, then slows

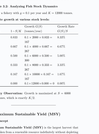

Figure 1. Logistic Growth

Example 1.1: Population with Carrying Capacity

A fish population follows the logistic equation

$$P(t) = \frac{1000}{1 + 9e^{-0.5t}}$$

where $P(t)$ is in tonnes and t is in years.

Questions: 1. What is the carrying capacity? 2. What is the initial population? 3. How long until the population reaches 50% of carrying capacity?

Solution

1. Carrying capacity: $K = 1000$ tonnes
2. Initial population:

$$P(0) = \frac{1000}{1 + 9e^{-0.5 \cdot 0}} = \frac{1000}{1 + 9} = 100 \text{ tonnes}$$

3. Time to reach $P = 500$ tonnes:

$$500 = \frac{1000}{1 + 9e^{-0.5t}}$$

$$1 + 9e^{-0.5t} = 2$$

$$9e^{-0.5t} = 1$$

$$e^{-0.5t} = \frac{1}{9}$$

$$-0.5t = \ln\left(\frac{1}{9}\right) = -\ln(9)$$

$$t = 2\ln(9) \approx 4.39 \text{ years}$$

Note: This matches Q6 in the exam, which asks about this time to reach 50% of carrying capacity.

Key Observations (Scroll to mentoment at $K = 1000$ tonnes, which is exactly $K/2$

2.3 The Schaefer Model of Fish Growth

From Population Trajectory to Growth Rate

The logistic function gives us population at time t. But the **growth function** $G(P)$ tells us how fast the population increases at any size.

The Schaefer Growth Equation

$$G(P) = rP\left(1 - \frac{P}{K}\right)$$

where: $G(P)$ = growth (flow) of fish biomass ρ = intrinsic growth rate (maximum growth when stock is small) P = current stock (or population) K = carrying capacity

Understanding the Model Components

The term $\left(1 - \frac{P}{K}\right)$ is the key:

Stock Level	Compensating Factor	Effect
$P \approx 0$	$\left(1 - \frac{P}{K}\right) \approx 1$	Growth near r	Stock near K	$\left(1 - \frac{P}{K}\right) \approx 0$	Growth near zero
$P = K/2$	$\left(1 - \frac{P}{K}\right) = 0.5$	Maximum growth
$P = K$	$\left(1 - \frac{P}{K}\right) = 0$	No growth
$P > K$	$\left(1 - \frac{P}{K}\right) < 0$	Negative growth (decline)

The Actual Growth Rate

The **actual growth rate** (growth per unit stock) is:

$$\frac{G(P)}{P} = r\left(1 - \frac{P}{K}\right)$$

This declines linearly from ρ (when $P \approx 0$) to 0 (when $P = K$).

Example 2.2: Analyzing Fish Stock Dynamics

Consider a fishery with $r = 0.4$ per year and $K = 2000$ tonnes.

Calculate growth at various stock levels:

Stock Level	Growth Rate	Actual Growth Rate
200	$0.4 \cdot 200 \cdot \left(1 - \frac{200}{2000}\right) = 72$	0.36
1000	$0.4 \cdot 1000 \cdot \left(1 - \frac{1000}{2000}\right) = 200$	0.20
1500	$0.4 \cdot 1500 \cdot \left(1 - \frac{1500}{2000}\right) = 150$	0.10
1800	$0.4 \cdot 1800 \cdot \left(1 - \frac{1800}{2000}\right) = 72$	0.04
2000	$0.4 \cdot 2000 \cdot \left(1 - \frac{2000}{2000}\right) = 0$	0.00
2200	$0.4 \cdot 2200 \cdot \left(1 - \frac{2200}{2000}\right) = -88$	−0.04

Key Observations (Scroll to mentoment at $K = 2000$ tonnes, which is exactly $K/2$

2.4 Maximum Sustainable Yield (MSY)

The Concept

Maximum Sustainable Yield (MSY) is the largest harvest that can be taken from a renewable resource indefinitely without depleting the stock.

Figure 2. Sustain Yield and MSY

At MSY: Harvest = Growth - Stock remains constant year after year

Finding the MSY

The Schaefer model $G(P) = rP\left(1 - \frac{P}{K}\right)$ is a **quadratic function** of P (a downward opening parabola).

To find the maximum, we can either:

- Use **calculus** (Week 5): $G'(P) = 0$
- Use **the vertex formula**: For $G(P) = aP^2 + bP$, the vertex is at $P = -\frac{b}{2a}$

Using the vertex formula:

Rewriting $G(P) = rP - \frac{r}{K}P^2$:

Standard form: $G(P) = aP^2 + bP$ where $a = -\frac{r}{K}$ and $b = r$

$$P_{MSY} = -\frac{b}{2a} = -\frac{r}{-\frac{2r}{K}} = \frac{K}{2}$$

MSY occurs when the stock is at half the carrying capacity:

$$P_{MSY} = \frac{K}{2}$$

The MSY Harvest is:

$$G_{MSY} = r \cdot \frac{K}{2}\left(1 - \frac{\frac{K}{2}}{K}\right) = \frac{rK}{2}\left(1 - \frac{1}{2}\right) = \frac{rK}{4}$$

Example 2.3: MSY Calculation

For $\rho = 0.5$ and $K = 1000$ tonnes:

$$P_{MSY} = \frac{1000}{2} = 500 \text{ tonnes}$$

$$G_{MSY} = \frac{0.5 \cdot 1000}{4} = 125 \text{ tonnes/year}$$

Interpretation: By maintaining the fish stock at 500 tonnes, we can sustainably harvest 500 tonnes every year - forever.

Management Implications

Stock Level	Management Action
$P < P_{MSY}$	Reduce harvest to let stock recover
$P = P_{MSY}$	Harvest exactly G_{MSY} to maintain
$P > P_{MSY}$	Can harvest more than G_{MSY} temporarily to reduce stock to optimal level

2.5 Connecting Modules: Exponential → Logistic

The Arithmetic Logistic Trick

The Schaefer model can also be written as a differential equation:

$$\frac{dP}{dt} = rP\left(1 - \frac{P}{K}\right)$$

When $P \approx K$, this simplifies to:

$$\frac{dP}{dt} \approx r(K - P)$$

which gives exponential growth $K(t) = K_0 e^{rt}$.

The relation to the full logistic differential equation is:

$$\frac{dP}{dt} = \frac{P}{K} \cdot r(K - P)$$

where P/K is a "correction factor" that slows growth as P approaches K.

The Schaefer growth model and the logistic function describe the same phenomenon - one focuses on growth rate, the other on the population trajectory.

2.6 Arithmetic Sequences: The Linear Analogue

Definition

While geometric sequences connect to exponential functions, **arithmetic sequences** connect to **linear functions**.

A sequence $\{a_n\}_{n=1}^{\infty}$ is **arithmetic** if consecutive terms differ by a constant:

$$a_n = a_1 + (n - 1)d$$

where: a_1 = first term d = **common difference** = a - term number

Example 2.4

Sequence: $\{3, 7, 11, 15, 19, \ldots\}$

$$a_1 = 3, \quad d = 4, \quad a_n = 3 + (n - 1) \cdot 4 = 4n - 1$$

$$\text{10th: } a_{10} = 39, \quad \text{100th: } a_{100} = 399, \quad \text{n: } a_n = 4n - 1$$

Connection to Linear Functions

An arithmetic sequence is **discrete sampling** of a linear function:

$$a_n = a_1 + (n - 1)d = f(n) = mx + b$$

Compare with slope-intercept form $y = mx + b$. Slope $m = d$ (the common difference). The sequence values lie exactly on the line at integer points.

Example 2.5: The Fish Stock Under Constant Decline

Suppose fish stock declines linearly due to environmental degradation:

$$S_n = 2000 - 100(n - 1)$$

where S_n is stock in year n.

Questions: 1. When is the initial stock? 2. What is the common difference? 3. When will the stock reach zero?

Solution

1. Initial stock: $S_1 = 2000 - 100(1 - 1) = 2000$ tonnes
2. Common difference: $d = -100$ tonnes/year (declining)
3. Stock reaches zero when:

$$2000 - 100(n - 1) = 0$$

$$n - 1 = 20$$

$$n = 21 \text{ years}$$

Arithmetic Series

The sum of the first n terms of an arithmetic sequence:

$$S_n = \sum_{k=1}^{n} a_k = \frac{n}{2}(a_1 + a_n) = \frac{n}{2}(2a_1 + (n - 1)d)$$

2.7 Function Composition

Definition

Given two functions f and g, the **composition** $f \circ g$ is defined as:

$$(f \circ g)(x) = f(g(x))$$

Read as: "f of g of x" - first apply g, then apply f to the result.

Important Notes

1. **Order matters:** Generally, $f \circ g \neq g \circ f$
2. **Domain restriction:** x must be in the domain of g, and $g(x)$ must be in the domain of f

Example 2.5: Composing Functions

Let $f(x) = x^2$ and $g(x) = 3x + 1$.

(a) Find $(f \circ g)(x)$:

$$(f \circ g)(x) = f(g(x)) = f(3x + 1) = (3x + 1)^2$$

(b) Find $(g \circ f)(x)$:

$$(g \circ f)(x) = g(f(x)) = g(x^2) = 3x^2 + 1$$

(c) Evaluate $(f \circ g)(2)$:

$$(f \circ g)(2) = (3 \cdot 2 + 1)^2 = 7^2 = 49$$

Example 2.6: Composition with Logistic Function

Let $f(x) = \frac{1}{1 + e^{-x}}$ (logistic function) and $g(x) = 2x$.

$$(f \circ g)(x) = f(2x) = \frac{1}{1 + e^{-2x}}$$

This is exactly the dimmer switch function from Week 1, Example 1.7!

Application: Building Complex Models

In a fish modeling, composition allows us to build complex models from simpler components:

$$\text{Rate} = f(\text{Biomass})$$

$$\text{Biomass} = g(\text{Time})$$

$$\text{Combined: Rate} = (f \circ g)(\text{Time})$$

2.8 Inverse Functions

Definition

If f is a one-to-one function, its **inverse** f^{-1} satisfies:

$$f^{-1}(f(x)) = x \quad \text{and} \quad f(f^{-1}(x)) = x$$

Graphically: The graph of f^{-1} is the reflection of the graph of f across the line $y = x$.

Finding Inverse Functions

Step 1: Write $y = f(x)$

Step 2: Solve for x in terms of y

Step 3: Swap x and y to get $y = f^{-1}(x)$

Example 2.7: Finding Inverses

(a) Find the inverse of $f(x) = 2x + 3$:

Step 1: $y = 2x + 3$

Step 2: Solve for x:

$$y = 2x + 3$$

$$x = \frac{y - 3}{2} = \frac{y}{2} - \frac{3}{2}$$

Step 3: Swap x and y:

$$f^{-1}(x) = \frac{x}{2} - \frac{3}{2}$$

Verification:

$$f(f^{-1}(x)) = f\left(\frac{x}{2} - \frac{3}{2}\right) = 2\left(\frac{x}{2} - \frac{3}{2}\right) + 3 = x \checkmark$$

(b) Find the inverse of $f(x) = x^2$:

Step 1: $y = x^2$

Step 2: Solve for x:

$$x = \pm\sqrt{y}$$

Step 3: $f^{-1}(x) = \sqrt{x}$

Inverse of the Logistic Function

For the logistic function $P = \frac{K}{1 + Ae^{-rt}}$, we can solve for t:

$$P(1 + Ae^{-rt}) = K$$

$$1 + Ae^{-rt} = \frac{K}{P}$$

$$Ae^{-rt} = \frac{K}{P} - 1$$

$$e^{-rt} = \frac{1}{A}\left(\frac{K}{P} - 1\right)$$

$$-rt = \ln\left(\frac{1}{A}\left(\frac{K}{P} - 1\right)\right)$$

$$t = -\frac{1}{r}\ln\left(\frac{1}{A}\left(\frac{K}{P} - 1\right)\right)$$

This is the **time required to reach a given population level**.

2.9 Why This Matters: Preparing for Optimization

Understanding bounded growth and the Schaefer model is crucial preparation for:

Week	Topic	Connection
Week 3: Derivatives	Finding $G'(P)$	Derivative of logistic function
Week 5: Optimization	Maximizing sustainable yield	Finding MSY mathematically
Week 6: Produce-Prey	Predator-prey models	Lotka-Volterra extends these ideas to two species
Week 10: Linear Programming	Resource allocation under constraints	Same logic: maximize subject to limits

The Schaefer model demonstrates how **quadratic structures** (from composition of $f(x) = rx(1 - x)$) lead to **optimization problems** (the vertex formula). Calculus will revisit this theme.

Summary: Key Formulas for Week 2

Topic	Key Formula
Logistic function	$P(t) = \frac{K}{1 + \frac{K - P_0}{P_0} e^{-rt}}$
Alternative form	$P(t) = \frac{K}{1 + e^{-r(t - t_0)}}$
Schaefer growth	$G(P) = rP\left(1 - \frac{P}{K}\right)$
Actual growth rate	$\frac{G(P)}{P} = r\left(1 - \frac{P}{K}\right)$
MSY stock level	$P_{MSY} = \frac{K}{2}$
MSY harvest	$G_{MSY} = \frac{rK}{4}$
Arithmetic sequence	$a_n = a_1 + (n - 1)d$
Arithmetic series	$S_n = \frac{n}{2}(2a_1 + (n - 1)d)$
Composition	$(f \circ g)(x) = f(g(x))$
Inverse relationship	$f^{-1}(f(x)) = x$

Looking Ahead: Week 3

In Week 3, we will formalize the concept of **instantaneous rate of change** through the **derivative**. This will allow us to:

- Find $G'(P)$ directly from the Schaefer model
- Understand why the vertex formula works for quadratic functions
- Continuous Limits and continuity as foundations for calculus

The optimization of the Schaefer model (Q5) is the comes, and by fully analytical using derivative techniques.

Materials adapted from NCEMS lesson notes (Mary Hardy) and aligned with IGCSE/AS/A Level Examinations.